

Emergency Food Network (EFN)

Service Demand, Utilization, and Distribution Analysis

Overview

This project provides an applied analysis of service activity within the Emergency Food Network (EFN) partner system using annual administrative data from 2014 to 2024. The analysis focuses on service reach, repeat utilization, food distribution capacity, demographic composition, and variation across service units.

The primary goal is descriptive: to characterize how recorded service activity changed over time and across the observed EFN network. The project does not attempt to estimate causal effects, evaluate program performance, or measure underlying food insecurity in the broader community.

The analysis is organized as a sequence of three notebooks. The first establishes the structure and limitations of the data, the second provides the core descriptive analysis, and the third adds a limited time-series extension to assess whether the aggregate annual service series supports a cautious baseline projection.

Project Structure

1. Data Structure, Definitions, and Validation

File: 01_Data_Structure_Definitions_and_Validation.pdf

This notebook establishes the analytical foundation of the project. It defines the unit of observation as a service unit-year record, clarifies the unbalanced panel structure, and distinguishes between ZIP-coded service units and program-level service identifiers.

It also documents the construction of total service encounters from unduplicated and duplicated counts, validates key accounting identities, and explains the interpretation of food flow variables. Special attention is given to zeros, missingness, and the analytical limits of the dataset.

2. Exploratory Data Analysis (EDA)

File: 02_Exploratory_Data_Analysis.pdf

This notebook provides the core descriptive analysis of EFN service activity. It examines dataset coverage, service reach over time, repeat utilization intensity, food distribution capacity, demographic composition, and variation across service units.

All results are descriptive in nature. The analysis is designed to identify patterns in service activity while avoiding causal claims or performance judgments.

3. Limited Time-Series Extension

File: Limited_Time_Series_Extension.pdf

This notebook evaluates whether the aggregate annual series of total service encounters supports a limited univariate time-series forecasting exercise.

Given the short annual sample and substantial instability after 2020, the emphasis is on transparency, parsimony, and uncertainty rather than high-confidence prediction. The selected baseline model is interpreted as a cautious planning extension rather than as the central contribution of the project.

Main Analytical Themes

- Service reach over time
- Repeat utilization and service intensity
- Food distribution capacity
- Demographic composition of clients served
- Variation across service units
- Limited time-series projection under uncertainty

Methodological Principles

- Descriptive, not causal
- Transparent data definitions and validation
- Careful interpretation of measurement limits
- Parsimonious time-series modeling
- Conservative treatment of uncertainty

Key Limitations

The project is constrained by the structure of the available data. The dataset records observed service activity within the EFN partner system rather than

underlying food insecurity in the broader community. The time-series extension relies on an annual series with only 11 observations, which limits model complexity and forecast precision. In addition, the post-2020 period is structurally unstable, reducing the reliability of trend extrapolation.

Tools

Python (`pandas` , `numpy` , `matplotlib` , `statsmodels`), Jupyter Notebooks, `nbconvert`

Author Note

This project was completed as an applied economics portfolio exercise using real administrative data. Interpretations are descriptive and intended to inform understanding of service activity, utilization, and uncertainty rather than policy evaluation or causal inference.